

ANDREW ROOT

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SUMMARY

Founding engineer, 2+ years in technical roles, who does not hand systems off at the door: I install them on customer sites, train the staff who use them, build the alerting that catches failures, and take the debugging call. I write Python across real-time audio systems, LLM APIs, and edge devices, and I build observability in from the start so I can troubleshoot what I ship.

TECHNICAL SKILLS

Languages	Python, C, SQL, TypeScript/JavaScript
AI	LLM APIs, LiveKit, STT, real-time audio agents, PyTorch, pose estimation
Infra & Observability	AWS (CDK, CloudWatch, SNS, SageMaker), Linux, Git/GitHub, Wi-Fi & network troubleshooting
Other	OpenCV, Open3D, edge devices, technical writing

PROFESSIONAL EXPERIENCE

Founding Engineer | *Daentra (early-stage startup)* | San Francisco, CA 08/25 - Present

- Deployed a real-time vision/audio system into dental operatory rooms across 2 customer offices, running every on-site installation personally and training clinical staff to use the system during live treatment; now operate 10 edge devices across 5 rooms.
- Cut failure detection from a customer phone call to an automatic alert by building the deployment's observability layer on AWS CloudWatch and SNS: structured, queryable logs, and alerting on device and pipeline faults.
- Took the Pico 2W off the customer's congested network after my alerting caught repeated device dropouts and I traced them to 2.4 GHz contention, reconfiguring each Orange Pi 5B as a dedicated access point for its paired Pico.
- Turned installation from undocumented practice into a repeatable checklist the team follows, authoring the two documents Daentra installs from: a deployment guide (supply lists, in-room installation) and a provisioning runbook (flashing, AWS device registration, pre-installation validation).
- Redesigned the in-room hardware install after the first office reported cabling interfering with treatment, rerouting wiring through the dental light and powering the Orange Pi, camera, and Pico from existing operatory circuits within the supply's ratings.

Research Lab Assistant | *Oakland University* | Oakland County, MI 05/24 - 05/25

- Developed a real-time computer vision pipeline (Python, PyTorch, OpenCV) that segments expected objects from live video for an automated security vehicle, built and evaluated with master's and PhD students, in collaboration with Dataspeed Inc.

Undergraduate Researcher | *University of Wisconsin Eau Claire* | Eau Claire, WI 09/23 - 12/24

- Quantified how far cyberbullying-detection models degrade on data they were never trained on, running cross-dataset evaluations (Python, XGBoost) on the university's Linux HPC cluster to isolate sources of dataset bias; mentored junior researchers in the research group.

Software Engineer Intern | *Midwest Manufacturing (Menards Inc.)* | Eau Claire, WI 05/23 - 05/24

- Put advanced SQL database operations in the hands of non-technical manufacturing teams by shipping a full-stack web application.
- Refactored 3 production manufacturing services integrating PLC control systems and message brokers.
- Prototyped a service that auto-generates Confluence documentation from code repositories via web scraping and syntax parsing.

PROJECTS & LEADERSHIP

Voice-Assistant Task Manager 🗣️ | Built a real-time voice agent on LiveKit that lets users manage tasks entirely by speech, with no UI controls: streams audio to a Python agent that transcribes it, resolves intent with an LLM, executes the matching database operations, and pushes updates to a Next.js frontend live. Diagnosed the agent's misfires by replaying session transcripts turn by turn against the actions it took.

President & Instructor, Students' Association for Computing Machinery | Led the university's ACM chapter and taught its technical program: designed and ran hands-on sessions on Python, Git/GitHub, statistics, and machine learning for 20+ students, and chaired a board of officers.

EDUCATION

B.S. Computer Science & Applied Math (Double Major) | UW Eau Claire (3.98/4.0), Honors 09/22 - 12/24